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Propensity-Matched 1-Year Outcomes Following Transcatheter Aortic Valve Replacement in Low-Risk Bicuspid and Tricuspid Patients

G. Michael Deeb, MD,[a] Michael J. Reardon, MD,[b] Basel Ramlawi, MD,[c,*] Steven J. Yakubov, MD,[d] Stan J. Chetcuti, MD,[e] Neal S. Kleiman, MD,[f] Abeel A. Mangi, MD,[g,†] Firas Zahr, MD,[h] Howard K. Song, MD, PHD,[i] Hemal Gada, MD, MBA,[j] Mubashir Mumtaz, MD,[k] John Heiser, MD,[l] William Merhi, DO,[m] C. Patrick Murrah, MD,[n] Thomas Noel, MD,[o] Merick Kirshner, MD,[p] Timothy Byrne, DO,[q] Saki Ito, MD, MSC,[r] Jian Huang, MD, MSC,[s] John K. Forrest, MD[t]

ABSTRACT

OBJECTIVES The aim of this study was to compare 1-year outcomes after transcatheter aortic valve replacement (TAVR) in low surgical risk patients with bicuspid aortic stenosis to patients with tricuspid aortic stenosis.

BACKGROUND The pivotal TAVR trials excluded patients with bicuspid aortic valves. The Low Risk Bicuspid Study 30-day primary endpoint of death or disabling stroke was 1.3%.

METHODS The Low Risk Bicuspid Study is a prospective, single-arm, TAVR trial that enrolled patients from 25 U.S. sites. A screening committee confirmed bicuspid anatomy and valve classification on computed tomography using the Sievers classification. Valve sizing was by annular measurements. An independent clinical events committee adjudicated all serious adverse events, and an independent core laboratory assessed all echocardiograms. The 150 patients from the Low Risk Bicuspid Study were propensity matched to the TAVR patients in the randomized Evolut Low Risk Trial using the 1:1 5- to-1-digit greedy method, resulting in 145 pairs.

RESULTS All-cause mortality or disabling stroke at 1 year was 1.4% in the bicuspid and 2.8% in the tricuspid group ($P = 0.413$). A pacemaker was implanted in 16.6% of bicuspid and 17.9% of tricuspid patients ($P = 0.741$). The effective orifice area was similar between groups at 1 year (2.2 ± 0.7 cm^2 vs 2.3 ± 0.6 cm^2, $P = 0.677$) as was the mean gradient (8.7 ± 3.9 mm Hg vs 8.5 ± 3.1 mm Hg, $P = 0.754$). Fewer patients in the bicuspid group had mild or worse paravalvular leak (21.3% vs 42.6%, $P < 0.001$).

CONCLUSIONS There were no significant differences in clinical or forward flow hemodynamic outcomes between the propensity-matched groups at 1 year. (J Am Coll Cardiol Intv 2022;15:511–522) © 2022 by the American College of Cardiology Foundation.

From the [a]Department of Cardiac Surgery, University of Michigan Hospitals, Ann Arbor, Michigan, USA; [b]Department of Cardiothoracic Surgery, Houston Methodist DeBakey Heart and Vascular Institute, Houston, Texas, USA; [c]Department of Cardiothoracic Surgery, Valley Health, Winchester, Virginia, USA; [d]Department of Interventional Cardiology, Riverside Methodist-OhioHealth, Columbus, Ohio, USA; [e]Department of Interventional Cardiology, University of Michigan Hospitals, Ann Arbor, Michigan, USA; [f]Department of Interventional Cardiology, Houston Methodist DeBakey Heart and Vascular Institute, Houston, Texas, USA; [g]Department of Surgery (Cardiac Surgery), Yale University School of Medicine, New Haven, Connecticut, USA; [h]Department of Interventional Cardiology, Oregon Health and Science University, Portland, Oregon, USA; [i]Department of Cardiothoracic Surgery, Oregon Health and Science University, Portland, Oregon, USA; [j]Department of Interventional Cardiology, University of Pittsburgh Pinnacle, Wormsleyburg, Pennsylvania; USA; [k]Department of Cardiothooracic Surgery, University of Pittsburgh Pinnacle, Wormsleyburg, Pennsylvania, USA; [l]Department of Cardiothoracic Surgery, Spectrum Health, Grand Rapids, Michigan, USA; [m]Department of Interventional Cardiology, Spectrum Health, Grand Rapids, Michigan, USA; [n]Department of Cardiothoracic Surgery, Tallahassee Research Institute, Tallahassee, Florida, USA; [o]Department of Interventional Cardiology, Tallahassee Research Institute, Tallahassee, Florida, USA; [p]Department of Cardiothoracic Surgery, Abrazo Arizona Heart Hospital, Phoenix, Arizona, USA; [q]Department of Interventional Cardiology, Abrazo Arizona Heart

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ABBREVIATIONS AND ACRONYMS

AS = aortic valve stenosis

BAV = bicuspid aortic valve

EOA = effective orifice area

LVOT = left ventricular outflow tract

KCCQ = Kansas City Cardiomyopathy Questionnaire

PPM = prosthesis-patient mismatch

PVL = paravalvular leak

SAVR = surgical aortic valve replacement

STS = Society of Thoracic Surgeons

TAVR = transcatheter aortic valve replacement

Randomized, prospective trials have shown that transcatheter aortic valve replacement (TAVR) has excellent short-term outcomes compared with surgery for severe acquired calcific tricuspid aortic valve stenosis regardless of the predicted risk of surgical mortality (1-6). Patients with bicuspid aortic valve (BAV) stenosis were excluded from these trials. BAV is a congenital anomaly that occurs in approximately 1% to 2% of the population and can present in multiple anatomic configurations (7-9). Patients with BAV present unique challenges for TAVR because of the morphologic variants in valve anatomical structure, annular and left ventricular outflow tract (LVOT) shape, dynamic flow patterns, and the amount and distribution of calcium deposition on the BAV (10,11).

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The American College of Cardiology/Society of Thoracic Surgeons (STS)/Transcatheter Valve Therapy Registry, which requires reporting of all commercial TAVR cases, includes many TAVR in BAV cases because the implanting sites determined the patients were not suitable surgical candidates. Retrospective studies have shown that TAVR in BAV can be performed with acceptable procedural success rates and good clinical outcomes but with higher rates of complications when propensity matched with concurrent tricuspid TAVR procedures (11-15). These registry-based trials were not controlled, and the events were not centrally adjudicated.

Promising early results in patients with bicuspid aortic valve stenosis (AS) at low surgical risk have been reported from 2 prospective, controlled trials: the investigator-initiated Low Risk TAVR trial (16) and the Medtronic Low Risk Bicuspid Study (17). This study reports TAVR outcomes between propensity-matched patients from the single-arm Low Risk Bicuspid Study (17) and tricuspid aortic valve stenosis patients from the TAVR arm from the randomized Evolut Low Risk Trial (4).

METHODS

STUDY DESIGN. The Low Risk Bicuspid Study is a multicenter, prospective, single-arm study that enrolled patients from December 2018 to October 2019 at 25 sites in the United States (Supplemental Table 1). These sites also participated in the multicenter, prospective, randomized Evolut Low Risk Trial, which enrolled patients from March 2016 to November 2018 and compared TAVR with surgery (SAVR) in low surgical risk tricuspid valve AS patients. The details of both studies have been published (4,17). For each study, the institutional review boards approved the study protocols, and each patient provided written, informed consent. The trials were conducted in accordance with the International Conference on Harmonization, Good Clinical Practice guidelines, and the Declaration of Helsinki. Follow-up is planned through 10 years. The decision to submit the manuscript was that of the first and senior authors (G.M.D. and J.K.F.).

STUDY PROCEDURES. Patient eligibility into the Low Risk Bicuspid Study and the Evolut Low Risk Trial was reviewed by the same national screening committee comprising interventional cardiologists and cardiovascular surgeons. Computed tomographic studies were reviewed to confirm appropriate access anatomy, valve sizing, and native aortic valve type. The bicuspid patients were categorized using the classification proposed by Sievers and Schmidtke (10). The same independent clinical events committee adjudicated all deaths and endpoint-related adverse events for both studies. All echocardiograms were assessed by the same independent core laboratory (Mayo Clinic, Rochester, Minnesota). Patients were assessed at baseline, discharge, 30 days, and 1 year and will be followed annually for 10 years.

PATIENTS. The same inclusion and exclusion criteria were used for both low-risk trials (Supplemental Table 2), except for differences specifically related to aortic valve morphology and patients in the Low-Risk Bicuspid study could be younger (>60 years)

Hospital, Phoenix, Arizona, USA; [r]Division of Cardiovascular Ultrasound, Mayo Clinic, Rochester, Minnesota, USA; [s]Department of Statistics, Medtronic, Minneapolis, Minnesota, USA; and the [t]Department of Internal Medicine (Cardiology), Yale University School of Medicine, New Haven, Connecticut, USA. *Dr Ramlawi is currently at the Department of Cardiothoracic Surgery, Lankenau Heart Institute, Wynnewood, Pennsylvania, USA. †Dr Mangi is currently at the Department of Cardiac Surgery, Medstar Washington Hospital Center, Washington, DC, USA.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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TABLE 1 Baseline Characteristics for the Unadjusted and Adjusted Patient Cohorts

	Unadjusted				Adjusted			
	Bicuspid (n = 150)	Tricuspid[a] (n = 700)	*P* Value	ASD	Bicuspid (n = 145)	Tricuspid (n = 145)	*P* Value	ASD
Age, y	70.3 ± 5.5	74.0 ± 5.8	<0.001	0.656	70.5 ± 5.5	71.2 ± 5.9	0.321	0.117
Body surface area, m^2	1.9 ± 0.2	2.0 ± 0.2	0.024	0.198	1.9 ± 0.2	2.0 ± 0.2	0.151	0.169
Female	72 (48.0)	264 (37.7)	0.019	0.209	68 (46.9)	71 (49.0)	0.724	0.041
STS score, %	1.4 ± 0.6	2.0 ± 0.7	<0.001	0.869	1.4 ± 0.6	1.7 ± 0.7	0.002	0.374
New York Heart Association functional class			0.064	NA			0.197	NA
I	3 (2.0)	74 (10.6)			3 (2.1)	10 (6.9)		
II	106 (70.7)	447 (63.9)			102 (70.3)	100 (69.0)		
III	40 (26.7)	178 (25.4)			39 (26.9)	35 (24.1)		
IV	1 (0.7)	1 (0.1)			1 (0.7)	0 (0.0)		
Diabetes mellitus	37 (24.7)	218 (31.1)	0.116	0.145	36 (24.8)	36 (24.8)	>0.999	0.000
Hypertension	112 (74.7)	592 (84.7)	0.003	0.251	109 (75.2)	114 (78.6)	0.486	0.082
Chronic lung disease/COPD	26 (17.7)	104 (15.5)	0.506	0.060	25 (17.6)	22 (15.5)	0.632	0.057
Peripheral arterial disease	14 (9.3)	49 (7.1)	0.339	0.083	13 (9.0)	4 (2.8)	0.043	0.262
Cerebrovascular disease	10 (6.7)	72 (10.3)	0.173	0.130	10 (6.9)	11 (7.6)	0.821	0.027
SYNTAX I score[b]	1.1 ± 2.9	1.9 ± 3.7	0.003	0.246	1.0 ± 2.6	1.3 ± 2.9	0.304	0.121
Previous CABG	2 (1.3)	16 (2.3)	0.754	0.072	2 (1.4)	3 (2.1)	>0.999	0.053
Previous PCI	11 (7.3)	99 (14.1)	0.024	0.221	11 (7.6)	15 (10.3)	0.411	0.098
Previous myocardial infarction	6 (4.0)	44 (6.3)	0.280	0.104	6 (4.1)	4 (2.8)	0.750	0.076
Atrial fibrillation or flutter	11 (7.3)	107 (15.4)	0.010	0.255	11 (7.6)	14 (9.7)	0.530	0.074
Pre-existing pacemaker	4 (2.7)	21 (3.0)	> 0.999	0.020	4 (2.8)	2 (1.4)	0.684	0.097
LV ejection fraction,[c] %	63.5 ± 8.3	61.7 ± 8.0	0.013	0.221	63.4 ± 8.3	61.9 ± 9.0	0.148	0.170
Mean AV area,[c] cm^2	0.8 ± 0.2	0.8 ± 0.2	0.030	0.204	0.8 ± 0.2	0.8 ± 0.2	0.015	0.291
Mean AV gradient,[c] mm Hg	49.9 ± 15.5	46.9 ± 12.2	0.030	0.211	49.7 ± 15.4	46.9 ± 11.8	0.084	0.204
Aortic annulus diameter, mm[d]	25.2 ± 2.5	24.9 ± 2.3	0.157	0.124	25.2 ± 2.5	25.1 ± 2.4	0.556	0.069
Total calcium volume, mm^3	855.3 ± 580.2	801.1 ± 559.2	0.286	0.095	849.3 ± 587.7	835.3 ± 589.5	0.840	0.029

Values are mean ± SD or n (%). [a]Patients implanted with a CoreValve bioprosthesis (Medtronic) were excluded from these analyses. [b]The SYNTAX (SYNergy between PCI with TAXUS and Cardiac Surgery) score is a measure of the severity and extent of coronary artery disease. [c]Site reported. [d]Perimeter derived.

ASD = absolute standardized difference; AV = aortic valve; BSA = body surface area; CABG = coronary artery bypass grafting; COPD = chronic obstructive pulmonary disease; ICD = implantable cardioverter-defibrillator; IPG = implanted pulse generator; LV = left ventricular; NA = not applicable; PCI = percutaneous coronary intervention; STS = Society of Thoracic Surgeons.

and needed to have an ascending aortic diameter ≤4.5 cm (4,17). Details of the screening procedure and patient selection for the Low Risk Bicuspid study have been published by Forrest et al (17). Of the total submitted study patients with confirmed bicuspid valve anatomy, 28 were excluded due to aorta/annular dimensions (ie, the relationship of sinus width to annulus dimensions), 1 for prohibitive LVOT calcification, and none were excluded for the amount of raphe calcification. We sought to compare outcomes after TAVR in matched cohorts of low-risk patients with bicuspid and tricuspid aortic valve stenosis, excluding any other anatomic restrictions, differences, or potential selection biases (eg, raphe calcification) that were not present in both studies.

Eligible patients had either severe symptomatic AS or met American Heart Association/American College of Cardiology Class IIa criteria for asymptomatic aortic valve replacement (18). A predicted risk of 30-day SAVR mortality <3.0% was required based on local multidisciplinary Heart Team assessment. Severe symptomatic AS was defined as an aortic valve area ≤1.0 cm^2 or an aortic valve area index ≤0.6 cm^2/m^2 or a mean gradient of at least 40 mm Hg or a maximum aortic valve velocity of at least 4.0 m/s by transthoracic echocardiography with symptoms of exertional dyspnea, syncope or presyncope, or angina (18). Patients in the Low Risk Bicuspid Study were propensity matched to patients with tricuspid aortic stenosis from the TAVR arm of the Evolut Low Risk Trial.

VALVE SELECTION AND IMPLANT TECHNIQUE. All patients in the Low Risk Bicuspid Study underwent TAVR with an Evolut R or Evolut PRO valve (Medtronic). The patients with tricuspid valve AS in the Evolut Low Risk Trial who underwent TAVR with an Evolut R or Evolut PRO valve were eligible for matching. Details of the implant procedure for each

TABLE 2 Procedural Outcomes for the Matched Cohort

	Bicuspid (n = 145)	Tricuspid (n = 145)	*P* Value
General anesthesia	93 (64.1)	87 (60.0)	0.468
Implanted valve size			<0.001
Evolut R 23 mm	0 (0.0)	2 (1.4)	
Evolut R 26 mm	1 (0.7)	30 (20.7)	
Evolut R 29 mm	1 (0.7)	31 (21.4)	
Evolut R 34 mm	60 (41.4)	60 (41.4)	
Evolut PRO 23 mm	0 (0.0)	0 (0.0)	
Evolut PRO 26 mm	30 (20.7)	7 (4.8)	
Evolut PRO 29 mm	53 (36.6)	15 (10.3)	
Preimplant balloon valvuloplasty	133 (91.7)	43 (29.7)	<0.001
Postimplant balloon dilation	53 (36.6)	45 (31.0)	0.321
Iliofemoral access	143 (98.6)	145 (100)	0.156
Embolic protection device	43 (29.7)	2 (1.4)	<0.001
Resheath or recapture performed	47 (32.4)	65 (44.8)	0.030
Mean implant depth at NCS,[a] mm	2.8 ± 1.7	3.9 ± 2.2	<0.001
Mean implant depth at LCS,[a] mm	4.2 ± 2.0	5.1 ± 2.3	<0.001
More than 1 valve implanted	5 (3.4)	2 (1.4)	0.447
Concomitant or staged PCI	4 (2.8)	3 (2.1)	>0.999
Device success	137/143 (95.8)	136/141 (96.5)	0.777
Absence of procedural mortality	143/144 (99.3)	145 (100)	0.498
Correct positioning of single valve in proper anatomical location	139/144 (96.5)	143/145 (98.6)	0.282
Absence of moderate or severe prosthetic regurgitation (at 18 h to 7 d)	143 (100)	138/141 (97.9)	0.121

Values are n (%), mean ± SD, or n/N (%). [a]By aortography.
LCS = left coronary sinus; NCS = non-coronary sinus; PCI = percutaneous coronary intervention.

study have been described (4,17). Pre-TAVR balloon dilation was strongly encouraged in the bicuspid study and left to the operator's discretion in the Evolut Low Risk Trial; annular sizing was prescribed for all patients.

ENDPOINTS. The primary outcome for these analyses is the composite of death, disabling stroke, or aortic valve-related hospitalization. Secondary outcomes include the composite of death or disabling stroke, death, stroke (disabling and nondisabling), repeat hospitalization for aortic valve disease, major bleeding, pacemaker implantation, prosthetic valve endocarditis, valve thrombosis, and valve-related dysfunction requiring a repeat procedure at 1 year. Hemodynamic performance (mean gradient, effective orifice area, prosthesis-patient mismatch [PPM], and the degree of paravalvular leak [PVL]) are also reported. Quality of life up to 1 year was assessed by New York Heart Association functional class and the Kansas City Cardiomyopathy Questionnaire (KCCQ). PPM was based on the Valve Academic Research Consortium 3 definition as follows: moderate PPM, an indexed effective orifice area (EOA) >0.65 and ≤0.85 cm^2/m^2 for patients with a body mass index (BMI) <30 kg/m^2 or >0.55 and ≤0.70 cm^2/m^2 for patients with a BMI ≥30 kg/m^2, and severe PPM, an indexed EOA ≤0.65 cm^2/m^2 for a BMI <30 kg/m^2 and ≤0.55 for a BMI ≥30 kg/m^2 (19). The composite of device success defined as the absence of procedural mortality, the correct position of a single valve in the proper anatomical position, and the absence of more than mild aortic regurgitation is also reported.

All patients underwent computed tomography before the procedure, which was centrally assessed (Medtronic). Comprehensive quantitative calcium analyses were derived from the semiautomated calcium scoring tool in 3mensio software (Research Version 8.1, Pie Medical). The segmentation threshold was set to the mode of Hounsfield units in the contrast-enhanced blood in the aortic root plus 200 Hounsfield units for each patient. The total calcium volume was the sum of the aortic root at the valve basal plane to top of the leaflets and the basal plane to 10 mm into the LVOT.

STATISTICAL METHODS. The primary analysis cohort was composed of patients who underwent an attempted transcatheter aortic valve implant. Echocardiographic outcomes are reported for patients with an implanted transcatheter valve. Continuous variables are reported as mean ± SD and were compared using the Student's *t*-test. Categoric variables are presented as frequencies and percentages.

TABLE 3 Outcomes at 30 Days for Matched Patient Cohort

	30 Days			1 Year		
	Bicuspid (n = 145)	Tricuspid (n = 145)	*P* Value	Bicuspid (n = 145)	Tricuspid (n = 145)	*P* Value
All-cause mortality, disabling stroke, or aortic valve-related rehospitalization	4 (2.8)	2 (1.4)	0.409	6 (4.1)	10 (6.9)	0.304
All-cause mortality or disabling stroke	2 (1.4)	1 (0.7)	0.565	2 (1.4)	4 (2.8)	0.413
All-cause mortality	1 (0.7)	0 (0.0)	0.317	1 (0.7)	3 (2.1)	0.317
Cardiovascular mortality	1 (0.7)	0 (0.0)	0.317	1 (0.7)	2 (1.4)	0.565
Any stroke	6 (4.1)	3 (2.1)	0.311	7 (4.8)	4 (2.8)	0.358
Disabling stroke	1 (0.7)	1 (0.7)	0.998	1 (0.7)	1 (0.7)	0.998
Nondisabling stroke	5 (3.5)	2 (1.4)	0.250	6 (4.2)	3 (2.1)	0.309
Myocardial infarction	1 (0.7)	0 (0.0)	0.317	2 (1.4)	3 (2.1)	0.649
Major bleeding	2 (1.4)	4 (2.8)	0.407	3 (2.1)	5 (3.5)	0.471
Acute kidney injury	3 (2.1)	2 (1.4)	0.654	3 (2.1)	2 (1.4)	0.654
Valve embolization or migration	2 (1.4)	1 (0.7)	0.562	2 (1.4)	1 (0.7)	0.562
Valve thrombosis	1 (0.7)	0 (0.0)	0.317	2 (1.4)	1 (0.7)	0.568
Valve endocarditis	0 (0.0)	0 (0.0)	NA	0 (0.0)	0 (0.0)	NA
Valve-related dysfunction requiring repeat procedure	0 (0.0)	0 (0.0)	NA	1 (0.7)	0 (0.0)	0.317
Permanent pacemaker implantation[a]	21 (14.5)	25 (17.2)	0.526	24 (16.6)	26 (17.9)	0.741
Permanent pacemaker implantation[b]	21 (14.9)	25 (17.5)	0.559	24 (17.1)	26 (18.2)	0.782
Aortic valve-related rehospitalization	3 (2.1)	1 (0.7)	0.314	5 (3.5)	7 (4.9)	0.560
Coronary artery obstruction	1 (0.7)	0 (0.0)	0.317	1 (0.7)	0 (0.0)	0.317
Conversion to open surgery	1 (0.7)	0 (0.0)	0.317	1 (0.7)	0 (0.0)	0.317
Mean AV gradient >20 mm Hg[c]	2 (1.4)	0 (0.0)	0.498	2 (1.6)	0 (0.0)	0.237
Prosthesis patient mismatch,[c,d] n	129	115	0.105	111	100	0.131
None	112 (86.8)	107 (93.0)		99 (89.2)	95 (95.0)	
Moderate	13 (10.1)	7 (6.1)		10 (9.0)	3 (3.0)	
Severe	4 (3.1)	1 (0.9)		2 (1.8)	2 (2.0)	

Values are number of patients with events (Kaplan-Meier estimate), except for the valve gradient and prosthesis-patient mismatch data, which are presented as the n (%). [a]Including patients with a pacemaker at baseline. [b]Excluding patients with a pacemaker at baseline. [c]Echocardiographic data provided for patients with a successful implant. [d]Per Valve Academic Research Consortium 3 (19).

AV = aortic valve; NA = not applicable.

Ordinal data were compared using the Cochran-Mantel-Haenszel test, and categoric data were compared using the Fisher exact test or the chi-square test. Propensity score matching (1:1) was performed using a 5- to-1-digit greedy match to patients with tricuspid aortic valve stenosis who underwent self-expanding TAVR in the Evolut Low Risk Trial. Matching was based on age (≤70 vs >70 years), sex, body surface area (≤1.9 m^2 vs >1.9 m^2), STS score (≤1.3% vs >1.3%), diabetes mellitus, cerebrovascular disease, atrial fibrillation or flutter, prior myocardial infarction, aortic annular perimeter (≤80 mm vs >80 mm), and total calcium volume (≤735 mm^3 vs >735 mm^3). The categorical cutoffs were based on the median values in the bicuspid patient cohort. This resulted in 145 matched pairs. Absolute standardized differences were calculated for the baseline characteristics to evaluate the balance before and after matching, with values <10% used to indicate no meaningful imbalance. Adverse events at 1 year are reported using the Kaplan-Meier estimator and compared using the log-rank test. All statistical analyses were performed using SAS software, version 9.4 (SAS Institute).

RESULTS

PATIENTS. A total of 150 patients underwent attempted TAVR in the bicuspid study and 700 patients underwent TAVR with an Evolut R or Evolut PRO valve in the Evolut Low Risk Trial. Propensity matching resulted in 145 matched pairs (**Figure 1**).

Before matching, patients with bicuspid AS compared with patients with tricuspid AS were younger (70.3 ± 5.5 years vs 74.0 ± 5.8 years, $P < 0.001$); more were female (48.0% vs 37.7%, $P = 0.024$); had a lower STS score (1.4% ± 0.6% vs 2.0% ± 0.7%, $P < 0.001$); had a lower SYNTAX (SYNergy between PCI with TAXUS and Cardiac Surgery) score (1.1 ± 2.9 vs 1.9 ± 3.7, $P = 0.003$); and

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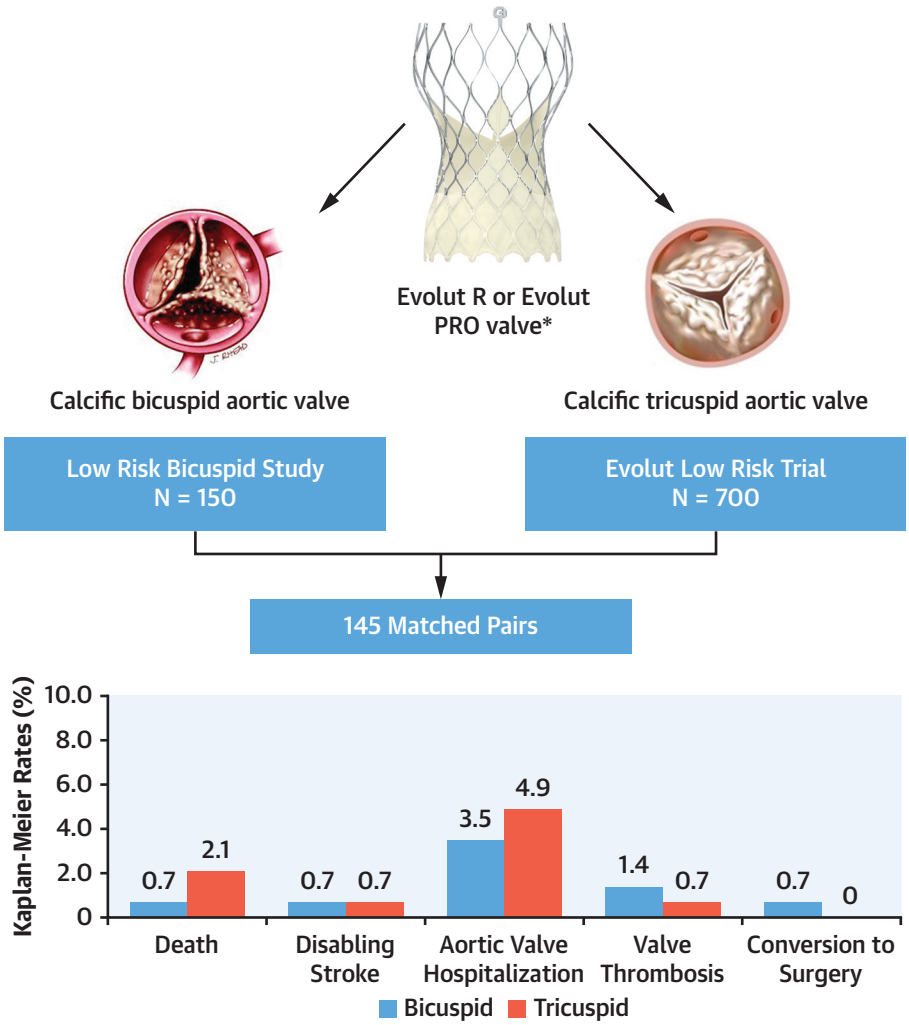
CENTRAL ILLUSTRATION Self-Expanding Transcatheter Aortic Valve Replacement in Patients With Bicuspid and Tricuspid Aortic Valve Stenosis



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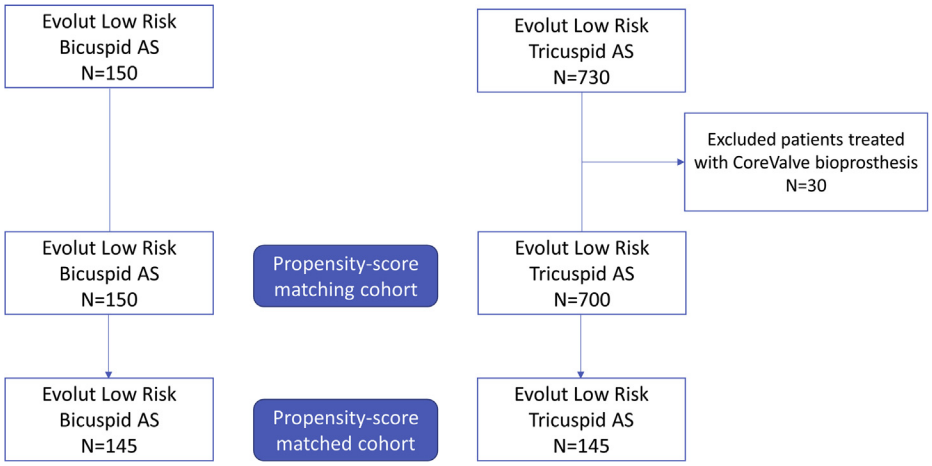
There were no significant differences in 1-year clinical outcomes between propensity-matched patients with bicuspid and tricuspid aortic valve stenosis. (Copyright Medtronic 2021)

fewer had prior percutaneous coronary interventions (7.3% vs 14.1%, $P = 0.024$), atrial fibrillation or flutter (7.3% vs 15.4%, $P = 0.010$), and hypertension (74.7% vs 84.7%, $P = 0.003$). Patients with bicuspid AS also had a higher left ventricular ejection fraction and higher mean aortic valve gradient at baseline (Table 1). After matching, the mean STS score was significantly lower in the bicuspid AS group (1.4% ± 0.6% vs 1.7% ± 0.7%, $P = 0.002$); however, the median scores were similar between groups (1.3% vs 1.4%, respectively). There were also more bicuspid patients who had peripheral arterial disease (9.0% vs 2.8%, $P = 0.043$). For the patients with bicuspid AS in the matched cohort, 14 (9.7%) had Sievers type 0 morphology, and 131 (90.3%) had Sievers type 1 morphology. Of the 131 patients with type 1 morphology, 103 patients (78.6%) had right-left fusion, 26 (19.8%) had right noncoronary fusion, and 2 (1.5%) had left noncoronary fusion. There were no patients with Sievers type 2.

PROCEDURAL OUTCOMES. Procedural outcomes for the matched patients are provided in Table 2. There were significant differences in the distribution of the implanted valve size between groups, but the 34-mm

FIGURE 1 Patients



All 150 patients in the Low Risk Bicuspid Study were propensity matched to patients in the Evolut Low Risk Trial who were implanted with similar valve types. This resulted in 145 matched pairs.

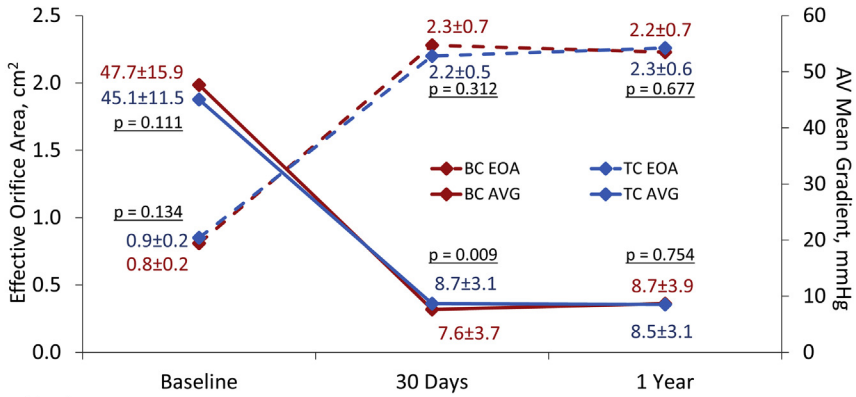
valve was used in the same number of patients in each group. Preimplant balloon valvuloplasty was performed significantly more often in the bicuspid AS group (91.7% vs 29.7%, $P < 0.001$). Also, there were more patients in the bicuspid group than in the tricuspid group in whom an embolic protection device was used (29.7% vs 1.4%, $P < 0.001$). The Evolut valves were resheathed or recaptured in fewer patients in the bicuspid group than in the tricuspid group (32.4% vs 44.8%, $P = 0.030$). The implant depth was significantly less in the patients in the bicuspid group; the noncoronary cusp depth was 2.8 ± 1.7 mm versus 3.9 ± 2.2 mm ($P < 0.001$), and the left coronary cusp depth was 4.2 ± 2.0 mm versus 5.1 ± 2.3 mm ($P < 0.001$). The device success rate was 95.8% in the bicuspid group and 96.5% in the tricuspid group ($P = 0.777$). A coronary artery occlusion occurred in 1 patient in the bicuspid group (0.7%), which resulted in conversion to surgery (0.7%), and there were no coronary occlusions or conversions in the tricuspid group ($P = 0.317$).

CLINICAL OUTCOMES AT 30 DAYS AND 1 YEAR. Clinical outcomes at 30 days for the 150 patients in the Low Risk Bicuspid Study have been published (17). Clinical outcomes at 30 days for the matched cohort are shown in **Table 3**. There were no significant differences between the 2 groups at 30 days. The median follow-up was 388.0 (Q1, Q3: 371.0,431.0) days in the bicuspid group and 733.0 (Q1, Q3: 553.0,748.0) days in the tricuspid group ($P < 0.001$).

One-year outcomes for all 150 patients are shown in Supplemental Table 3. The clinical outcomes at 1 year for the matched cohort are shown in **Table 3** and the **Central Illustration**. There were no differences in any serious adverse event rates between the patients in the bicuspid and tricuspid groups. From 30 days to 1 year, there were no additional deaths or disabling strokes in the bicuspid group, and 3 additional deaths in the tricuspid group for a composite 1-year rate of 1.4% versus 2.8% ($P = 0.413$). The composite of all-cause mortality, disabling stroke, or aortic valve-related hospitalization was 4.1% versus 6.9% ($P = 0.304$). All-cause mortality rates were 0.7% in the bicuspid group and 2.1% in the tricuspid group ($P = 0.317$). There were no disabling strokes in either group after 30 days. The rate of a new permanent pacemaker was 17.1% in the bicuspid group and 18.2% in the tricuspid group ($P = 0.782$). There was no difference in the aortic valve-related hospitalizations between groups (5 [3.5%] vs 7 [4.9%], $P = 0.560$).

Valve thrombosis was identified in 2 patients in the bicuspid group at 1 year (1.4%). The first patient had a spontaneous myocardial infarction 10 days post-TAVR and developed heparin-induced thrombocytopenia and died (17). The second patient was implanted with 2 Evolut PRO 29-mm valves at the index procedure after the first valve embolized to just

FIGURE 2 Valve Hemodynamics for the Matched Patient Cohort



No. of patients with echos	Baseline	30 Days	1 Year
Bicuspid EOA	131	129	113
Tricuspid EOA	120	115	101
Bicuspid AVG	144	142	125
Tricuspid AVG	143	141	131

The mean effective orifice area (EOA) and the mean aortic valve gradient (AVG) at 30 days and 1 year for patients in the bicuspid and tricuspid groups. All echocardiographic results are based on independent core laboratory assessment. The ± values are mean ± SD. The *P* value is based on the Fisher exact or chi-square test. AV = aortic valve.

above the coronary ostia at delivery catheter removal. At 199 days post-TAVR, restricted movement and thickening of the right coronary TAV leaflet was observed on echocardiography. The patient was hospitalized and treated with anticoagulation and was discharged with a peak velocity of 2.3 m/s. At 1 year, there was 1 patient in the tricuspid group who had valve thrombosis (0.7%) in the noncoronary sinus of Valsalva as seen on computed tomographic imaging, which was treated medically.

VALVE PERFORMANCE. The mean EOA and mean aortic valve gradients over time are shown in Figure 2, which shows similar mean EOAs between groups at 30 days and 1 year. The mean gradient at 30 days was significantly lower in the bicuspid group compared with the tricuspid group (7.6 vs 8.7 mm Hg, $P = 0.009$) but not at 1 year (8.7 vs 8.5 mm Hg, $P = 0.754$). There were 2 patients in the bicuspid group who had mean aortic valve gradients >20.0 mm Hg postprocedure (1.6%), which were still present at 1 year, and no patients in the tricuspid group. In patients with PVL measurements at both 30 days and 1 year, there were significantly fewer patients in the bicuspid group compared with the tricuspid group with mild or more PVL (21.3% vs 42.6%, $P < 0.001$) at 1 year but no difference in the proportion of patients with moderate or severe PVL between groups at 1 year (0.8% vs 3.9%, $P = 0.214$) (Figure 3). There was no difference in PPM between the 2 groups, and only 2 patients in each group had severe PPM at 1 year (Table 3).

QUALITY OF LIFE. There were no differences between groups in the mean KCCQ overall summary score for patients in the bicuspid versus tricuspid groups at 30 days (90.3 ± 12.6 vs 88.6 ± 14.4, $P = 0.295$) or at 1 year (92.8 ± 11.7 vs 91.0 ± 12.9, $P = 0.212$). There were also no differences between groups in the change in KCCQ score from baseline to 1 year (23.9 ± 20.0 vs 21.2 ± 22.3, $P = 0.303$). There were also no differences in the proportion of patients who had improvement in New York Heart Association functional class from baseline to 1 year between the bicuspid and tricuspid groups (84.4% vs 80.9%, $P = 0.371$).

DISCUSSION

Bicuspid aortic valves present unique challenges for TAVR due to morphologic variants in valve structure, annular and LVOT shape, dynamic flow patterns, and amount and distribution of calcium deposition on the BAV (10,11). Because of these differences, there were

legitimate concerns about procedural malposition of the device as well as mortality and morbidity with complications such as aortic tear, root rupture, coronary artery occlusion, and conversion to open surgery using a standard TAVR device developed for tricuspid anatomy to treat severe aortic stenosis in patients with BAV. Additionally, concerns existed over significant postprocedural mortality, stroke, aortic regurgitation, and high residual gradients.

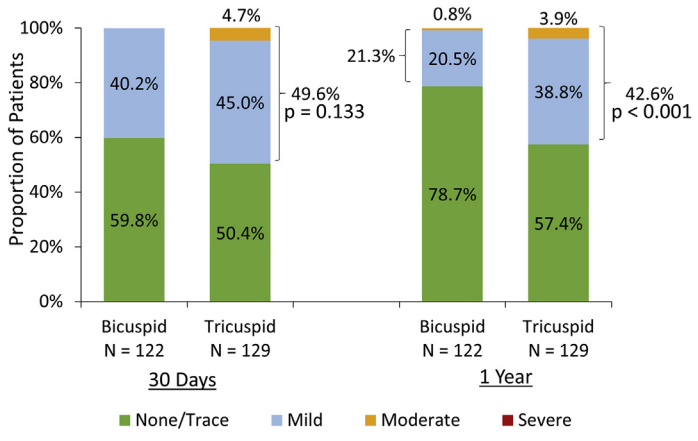
Despite these concerns, in this retrospective propensity score-matched comparison of low-risk patients with bicuspid versus tricuspid AS undergoing TAVR with a supra-annular, self-expanding valve, the device success rate was similar in both groups (95.8% vs 96.5%, $P = 0.777$). In addition, the Evolut valves were resheathed or recaptured in significantly fewer patients in the bicuspid group versus the tricuspid group (32.4% vs 44.8%, $P = 0.030$), and there were few procedural complications (0.7% vs 0.0%, $P = 0.317$).

Also, we found no significant differences in 30-day and 1-year clinical outcomes between groups. There were no clinically significant differences in forward flow hemodynamics or in the presence of PPM at 1 year; however, patients in the bicuspid group had significantly less mild or worse PVL at 1 year (21.3% vs 42.6%, $P < 0.001$). Preimplant balloon aortic valvuloplasty was performed significantly more often in the bicuspid study, secondary to strong recommendation from the principal investigators, due to its unique anatomic structure. Pre-TAVR balloon aortic valvuloplasty in the patients in the tricuspid group was left to the operators' discretion. Whether the difference in outcomes and severity of postprocedural PVL between the 2 groups was affected by the significantly higher number of pre-TAVR balloon valvuloplasty procedures performed in the bicuspid patients is uncertain; however, in standard practice for both studies, all investigational sites would perform a postprocedure balloon valvuloplasty if the initial postprocedure angiogram or echocardiogram showed mild or greater PVL.

This study is 1 of 2 prospective, controlled multicenter trials that previously reported excellent 30-day clinical and hemodynamic outcomes using TAVR to treat bicuspid aortic valve severe symptomatic and critical asymptomatic AS in low-risk patients (16,17). Our current 1-year outcomes in patients with bicuspid AS are excellent, with a composite of all-cause mortality or disabling stroke of 1.4%. There was 1 death (cardiovascular) in the bicuspid group at 1 year (0.7%) and 1 disabling stroke (0.7%). Valve dysfunction requiring repeat procedure (0.7%) and aortic valve-related rehospitalization (3.5%) were minimal. Forward flow hemodynamics were superb with a single-digit mean valve gradient of 8.6 mm Hg and an EOA of 2.2 cm^2. PVL, which was a concern using a device for bicuspid anatomy that was designed for a tricuspid valve and LVOT anatomy, was exceptional, with only 1 patient (0.8%) having more than mild PVL at 1 year. Our 1-year results are similar to those reported in the bicuspid patients from the controlled Low Risk TAVR trial (20), which reported mortality rates of 1.5% in the bicuspid patients and 3.0% in the tricuspid patients. Unlike our study, Rogers (20) included patients implanted with both self-expanding and balloon-expandable valves, and patients were enrolled in their low-risk bicuspid registry simultaneously with those in the low-risk randomized TAVR versus SAVR trial.

Retrospective uncontrolled studies comparing patients with bicuspid versus tricuspid AS undergoing TAVR from the Transcatheter Valve Therapy Registry have shown conflicting but acceptable results with newer generational devices (12-15). Yoon et al (15) showed that patients with bicuspid aortic valves underwent more frequent conversion to surgery (2.0% vs 0.2%, $P < 0.006$) and had a significantly lower device success rate (85.3% vs 91.4%, $P < 0.002$). Yoon et al (15) and Halim et al (13) have also shown differences in outcomes based on the generation of the device, with improved outcomes in BAV with the

FIGURE 3 PVL for the Matched Patient Cohort



The paravalvular leak (PVL) at 30 days and 1 year for the patients in the bicuspid and tricuspid groups. Data presented for patients who had PVL measured at both time points. Mild or more PVL was compared using the Fisher exact or chi-square test.

newer generation valves. Makkar et al (14) showed a significantly higher risk for stroke and procedural complications in patients with BAV requiring conversion to surgery.

Our reported results differ from previously reported uncontrolled studies in the literature due to study design (11-15) and the patients in our study had a lower mean STS score of <2.0%. The Low Risk Bicuspid Study, although a single-arm trial, was designed to ensure patients were implanted at the same centers as the Evolut tricuspid low-risk randomized TAVR versus surgery trial using the same central adjudication review of events procedure and the same echocardiographic core laboratory. All patients, including computed tomographic scans, were assessed by the same screening committee allowing consistent valve sizing using the annular perimeter at the basilar plane (11) and consistent root and valve anatomical criteria to differentiate bicuspid versus tricuspid anatomy for study inclusion and exclusion as well as verify that the patients met all other criteria set forth for the trials. All sites used the most advanced versions of the same valve. These matched controlled steps between the 2 trials minimized variables affecting outcomes and enhanced the impact of the bicuspid anatomy alone on the outcomes.

Our analyses from these 2 prospective controlled trials using experienced sites and third-generation supra-annular self-expanding valves show that TAVR in low-risk patients with AS is safe and effective at 1 year irrespective of valve anatomy. The long-term 10-year data from this single-arm Low Risk Bicuspid Study will be essential to determine TAVR durability, the need for reintervention or surgical crossover, and the survival rate for this patient population. Ideally, a 10-year randomized prospective multicenter trial comparing clinical and hemodynamic outcomes for TAVR versus surgery in low surgical risk bicuspid AS patients would provide long-term data for evidence-based decision making when determining procedure of choice in this patient population with respect to lifetime management of aortic valve disease.

STUDY LIMITATIONS. The Low Risk Bicuspid Study was performed at 25 sites who were also participating in the randomized Evolut Low Risk Trial and thus had significant experience with TAVR using this third-generation self-expanding TAVR valve. Data have shown that TAVR outcomes are related to procedural volume (21), and as such these data cannot necessarily be extrapolated to all TAVR centers. Because we were comparing TAVR between 2 patient groups with histologically different valve anatomy, a prospective randomized trial was not feasible; therefore, propensity score matching was used to control for differences in patient characteristics between the 2 patient groups. In addition, there were statistically significant differences in 2 baseline characteristics (ie, the mean STS score and the presence of peripheral arterial disease); yet, these may or may not be clinically meaningful. The propensity score matching was based on an STS score as a categoric variable (≤1.3% vs >1.3%), so the significant differences in the mean STS score after matching may not be clinically meaningful because the median STS scores for the 2 groups were 1.3% and 1.4%. The presence of peripheral arterial disease was not used in the propensity matching, and the differences may be due to chance. Although sites in the Low Risk Bicuspid Study were encouraged to enroll all patients with bicuspid valve disease requiring aortic valve replacement who met inclusion criteria, the potential for selection bias is present. The patients in the trials presented here were selected as suitable TAVR by a central screening committee and may not represent real-world clinical practice. Even though the Evolut Low Risk Trial showed that TAVR was noninferior to surgery and we have shown that low-risk patients with bicuspid AS had similar outcomes to patients with tricuspid AS after TAVR, any extrapolation of TAVR in BAV to surgical outcomes should be avoided.

CONCLUSIONS

There were no significant differences in any procedural, clinical, or forward flow hemodynamic outcomes for the propensity-matched groups at 30 days or 1 year. The Evolut supra-annular valve was highly effective at resolving severe symptomatic AS in patients with both bicuspid aortic valve anatomy reflecting the natural occurrence of Sievers types and tricuspid aortic valve anatomy.

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ADDRESS FOR CORRESPONDENCE: Dr G. Michael Deeb, Department of Cardiac Surgery, University of Michigan Hospitals, 1500 East Medical Center Drive, Ann Arbor, Michigan 48109-5864, USA. E-mail: mdeeb@med.umich.edu. Twitter: @johnkforrest.

PERSPECTIVES

WHAT IS KNOWN? The procedural safety, efficacy, and clinical and hemodynamic outcomes for TAVR therapy in patients with severe tricuspid aortic valve stenosis irrespective of patient surgical risk status has been demonstrated and is now an accepted modality of treatment. Previous registry studies have shown variable results for procedural, clinical, and hemodynamic outcomes using TAVR for treatment in patients with severe aortic valve stenosis and bicuspid aortic valve anatomy.

WHAT IS NEW? This propensity-matched study between 2 similarly designed prospective clinical trials compares the procedural and 1-year clinical and hemodynamic outcomes in patients with tricuspid or bicuspid severe aortic valve stenosis and demonstrates that low surgical risk patients with severe aortic stenosis irrespective of valve anatomy have similar outcomes when receiving a supra-annular self-expanding TAVR valve.

WHAT IS NEXT? Although the 1-year data for procedural, clinical, and hemodynamic outcomes in the single-arm Low Risk Bicuspid Study are excellent, the planned 10-year follow-up is essential to determine the long-term durability, rate of reintervention, or crossover to surgery and importantly all-cause mortality in this patient population.

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KEY WORDS aortic stenosis, bicuspid aortic valve, supra-annular, transcatheter aortic valve replacement

APPENDIX For supplemental tables, please see the online version of this paper.